

Beyond QKV : Minimal and Dynamic Projection Space Bottlenecks in Transformers

A Research Plan for ICLR

Research Proposal

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Abstract

Recent advances in efficient attention mechanisms, notably Keyless Attention, demonstrate that standard Key projections (W_K) can be eliminated by routing queries directly through a static transformation in Value space. However, foundational questions remain regarding the expressivity bounds, head interaction models, and extreme limits of projection reduction. This document presents a comprehensive, structured research plan detailing three high-impact tracks: (1) Amortized/Dynamic Keyless Attention, (2) Multi-Head Joint Routing via Cross-Head Transformations, and (3) Extreme Single-Projection Attention.

1 Introduction & Overall Objectives

Standard Multi-Head Attention (MHA) relies on three sequence-cached projection matrices ($W_Q, W_K, W_V \in \mathbb{R}^{d \times d_h}$). While effective, storing both K and V tensors during decoding imposes severe memory bandwidth bottlenecks. Keyless Attention proves that a static routing matrix $W_R \in \mathbb{R}^{d_h \times d_v}$ can bridge Q and V , reducing Key-Value (KV) cache storage by 50%.

This project explores the theoretical limits and practical enhancements of value-space routing across three complementary research tracks:

1. **Track 1:** Dynamic, context-conditioned routing transformations $W_R(x_i)$ to recover expressivity on complex sequential tasks.
2. **Track 2:** Cross-head value-routing matrices W_R^{global} to eliminate inter-head structural redundancy.
3. **Track 4:** Extreme projection compression, proving the bounds of single-projection attention ($Z = XW_Z$).

2 Track 1: Amortized Keyless Attention (Dynamic Context-Conditioned W_R)

2.1 Motivation & Mathematical Formulation

Static Keyless Attention employs a constant matrix $W_R \in \mathbb{R}^{d_h \times d_v}$ per head. While memory-efficient, W_R cannot adapt its subspace alignment dynamically based on input context. We propose parameterizing W_R as a rank- r dynamic operator amortized over the query state $q_i = x_i W_Q$:

$$W_R(x_i) = U_R \cdot \text{diag}(g(x_i W_Q)) \cdot V_R^\top \quad (1)$$

where $U_R \in \mathbb{R}^{d_h \times r}$, $V_R \in \mathbb{R}^{d_v \times r}$, and $g : \mathbb{R}^{d_h} \rightarrow \mathbb{R}^r$ is a lightweight linear function or two-layer MLP. The pre-softmax logit score between query token i and keyless token j becomes:

$$S_{i,j} = \frac{1}{\sqrt{d_h}} (x_i W_Q) W_R(x_i) (x_j W_V)^\top \quad (2)$$

2.2 Hypotheses

Hypothesis 1 (H1.1). *Static W_R models suffer performance degradation on multi-hop associative recall and long-context needle-in-a-haystack tasks due to static subspace alignment constraints.*

Hypothesis 2 (H1.2). *Dynamic rank- r parameterization $W_R(x_i)$ recovers full QKV routing expressivity while maintaining a strict 100% Value-Only Cache footprint.*

2.3 Experimental Execution

- **Synthetics:** Passkey retrieval, induction head formation, multi-query associative recall.
- **Pre-training:** 350M and 1.3B models trained on 50B tokens of SlimPajama.
- **Baselines:** Standard QKV MHA, Static Keyless Attention, Multi-Query Attention (MQA), Grouped-Query Attention (GQA).

2.4 Preliminary Results

A small-scale synthetic pilot (`keyless_tasks.py`: 2-layer, 128-dim encoders, single seed) tests H1.1/H1.2 directly before committing to the 350M/1.3B pre-training sweep above. Two tasks were implemented: associative recall (unique key–value pairs, then a repeated-key query; standard Based/Zoology-style diagnostic) and induction heads (a bigram recurs later in the sequence). The induction task did not reach a trustworthy baseline – even the full QKV model plateaus around 41% regardless of model size, data, or training budget, most likely because induction-head formation is normally studied in causal decoder-only models and does not transfer cleanly to this synthetic harness’s non-causal, single-scored-position encoder; it needs a causal reformulation before its numbers mean anything and is excluded below.

Associative recall gave a clean, trustworthy result (QKV reaches near-ceiling, confirming the task itself is well-posed):

Variant	Accuracy
QKV (baseline)	0.9995
Static Keyless	0.350
Dynamic Keyless (rank 8–64)	0.348–0.363

Table 1: Associative recall, 2-layer/128-dim encoder, single seed.

H1.1 is confirmed: static keyless collapses relative to QKV on this content-addressed task. **H1.2, as formulated, is not supported:** sweeping the dynamic operator’s rank from 8 to 64, and separately sweeping depth from 2 to 4 layers, left accuracy flat at ≈ 0.35 – 0.37 for both keyless variants – ruling out both capacity and compositional depth as the limiting factor for $W_R(x_i) = U_R \text{diag}(g(x_i W_Q)) V_R^\top$ specifically.

A follow-up diagnostic (not part of the original plan, added to explain the above) isolates why: both keyless variants always route the attention score through V itself (S_{ij} is some function of v_j in every case), so addressing and payload are never independently representable, no matter how W_R is parameterized. Replacing the score computation with a genuinely independent (but still low-rank, hence still far cheaper to cache than a full K) key projection produces a

sharp transition: rank ≤ 20 stays near keyless’s ≈ 0.35 – 0.40 ceiling, while rank ≥ 24 recovers near- QKV accuracy (0.99+) at most seeds tried, with some run-to-run inconsistency exactly at the boundary (rank 30 dropped back to 0.46 in one run) consistent with an optimization phase transition – the model either discovers the right circuit or gets stuck on a shortcut – rather than a smooth capacity curve; this boundary was checked at a single seed per rank and needs multi-seed confirmation before being taken as precise. The qualitative conclusion is nonetheless clear: *independence from V* , not merely more parameters or depth, is what static/dynamic keyless attention is missing.

This has a direct theoretical explanation in Gao & Xu’s own paper (“Keyless Attention: Value-Space Routing and Value-Only Caching for Efficient Transformers,” arXiv:2606.21848), which this project builds on. Their Theorem 1 proves that *single-head* keyless attention is an exact reparameterization of standard attention whenever W^V is invertible ($\tilde{W}^Q = W^Q(W^K)^\top(W^V)^{-\top}$) – no expressivity loss in principle. Theorem 2 extends this to the multi-head case, but only under an additional subspace condition, $\text{col}(\Omega_h^\top) \subseteq \text{col}(W_h^V)$, that is not automatically satisfied by training. We tested this directly on associative recall: single-head ($N_h=1$) static keyless attention reaches 1.0000 accuracy, exactly matching QKV (1.0000), across 3 seeds with zero exceptions – a clean empirical confirmation of Theorem 1. The identical model with $N_h=4$ heads (same total parameters, just split across heads) collapses to 0.34–0.37 across the same 3 seeds, while QKV at $N_h=4$ still solves it (0.9995+). The failure mode identified above is therefore not a defect of keyless attention per se, but specifically of *multi-head* keyless attention failing to satisfy Theorem 2’s subspace condition in practice – our independent-low-rank- K diagnostic is one way to sidestep that condition entirely (by not needing it to hold), but a more principled fix suggested directly by their own theory would encourage or enforce $\text{col}(\Omega_h^\top) \subseteq \text{col}(W_h^V)$ per head (e.g. via an explicit regularizer, a head-specific reparameterization, or an initialization scheme), rather than abandoning the value-only-cache property with an independent K .

This reframes Track 1 with a testable, theory-grounded target: a dynamic $W_R(x_i)$ (or any keyless variant) should be judged by whether it helps satisfy Theorem 2’s per-head subspace condition, not simply by adding capacity or query-conditioning in the abstract – which is exactly what $W_R(x_i) = U_R \text{diag}(g(x_i W_Q)) V_R^\top$ (H1.2 as originally stated) does not do, and why it did not help. H1.2 should be revised accordingly before further compute is committed to the full pre-training sweep.

3 Track 2: Multi-Head Joint Routing via Cross-Head W_R

3.1 Motivation & Mathematical Formulation

Standard MHA isolates attention heads prior to the final output projection (W_O). We extend value-space routing across heads by defining a global cross-head routing operator. Let $Q_{\text{concat}} = [Q^{(1)}, \dots, Q^{(H)}] \in \mathbb{R}^{N \times (Hd_h)}$ and $V_{\text{concat}} = [V^{(1)}, \dots, V^{(H)}] \in \mathbb{R}^{N \times (Hd_v)}$.

The global scoring matrix is computed as:

$$S_{\text{global}} = Q_{\text{concat}} W_R^{\text{global}} V_{\text{concat}}^\top, \quad \text{where } W_R^{\text{global}} \in \mathbb{R}^{(Hd_h) \times (Hd_v)} \quad (3)$$

To maintain efficiency, we factorize W_R^{global} into block-diagonal and low-rank cross-head components:

$$W_R^{\text{global}} = \text{BlockDiag}\left(W_R^{(1)}, \dots, W_R^{(H)}\right) + U_{\text{cross}} V_{\text{cross}}^\top \quad (4)$$

where $U_{\text{cross}} \in \mathbb{R}^{(Hd_h) \times r_{\text{global}}}$ and $V_{\text{cross}} \in \mathbb{R}^{(Hd_v) \times r_{\text{global}}}$.

3.2 Hypotheses

Hypothesis 3 (H2.1). *Cross-head value routing mitigates head redundancy, allowing lower-dimensional head projections to collaboratively learn non-orthogonal attention maps.*

Hypothesis 4 (H2.2). *Low-rank cross-head interaction serves as a hardware-friendly alternative to DeepSeek-style Multi-Head Latent Attention (MLA) without explicit latent tensor compression.*

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4 Track 4: Extreme Projection Compression (Single-Projection Attention)

4.1 Motivation & Mathematical Formulation

We push projection reduction to its theoretical limit by asking: *Can a Transformer operate using only one sequence-cached hidden representation?*

Let $Z = XW_Z \in \mathbb{R}^{N \times d_z}$ be the single sequence-cached state. Query and Value representations are derived via static parameter adapters $R_Q \in \mathbb{R}^{d_z \times d_h}$ and $R_V \in \mathbb{R}^{d_z \times d_v}$:

$$Q = ZR_Q = XW_ZR_Q, \quad V = ZR_V = XW_ZR_V \quad (5)$$

The single-cache attention score is given by:

$$S_{\text{single}} = (XW_ZR_Q)W_R(XW_ZR_V)^\top = X \left(W_ZR_QW_RR_V^\top W_Z^\top \right) X^\top \quad (6)$$

4.2 Theoretical Foundations & Rank Theorem

Theorem 1 (Rank Bottleneck Theorem). *Let $M_1 = W_ZR_QW_RR_V^\top W_Z^\top \in \mathbb{R}^{d \times d}$ be the single-projection attention score matrix. The rank of M_1 is bounded by:*

$$\text{rank}(M_1) \leq \min(d_z, d_h, d_v) \quad (7)$$

Furthermore, if $d_z \geq d_h + d_v$, single-projection attention can express any bilinear attention topology capable of being represented by 2-projection Keyless Attention.

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5 Milestones & Roadmap

Table 2: 4-Month Execution Timeline for ICLR Submission

Period	Key Milestones & Deliverables
Month 1	Formalize LaTeX proofs for Theorem 1; write Triton/PyTorch prototypes for Tracks 1, 2, and 4.
Month 2	Run synthetic diagnostic benchmarks (Passkey, Associative Recall); sweep hyperparameter ranks r and r_{global} .
Month 3	Execute 50B token pre-training sweeps on 350M / 1.3B models using SlimPajama; evaluate downstream zero-shot accuracy.
Month 4	Profile VRAM savings and decode throughput with custom Triton kernels; finalize paper draft and release code.